OpticNeuroBlender: auditable coherent scalar learning from evaluated Blender geometry

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Abstract

We implement a coherent scalar optical classifier whose fields and trainable phases originate from evaluated Blender geometry. An independent exact auditor admits a 133-state graph representing every nonzero branch of a 104-object scene and 17,060 terminal path occurrences. Captured-triangle GPU selection and FP64 graphical optical derivatives drive 60 own optimizer updates, with 61 actual Blender geometry captures and a saved/reopened final scene. CPU topology, affine origin jets, coherent merging, loss and Adam remain explicit. Independent rational analysis encloses all 150 final observed fields and powers, including train-only encoding, and separates numerical decisions from classification errors. A previously used Iris split yields 27/30 held-out predictions. Prospectively fixed Wine experiments yield 33/37, 30/37 and 28/37 against 32/37 matched baselines, without established superiority. Actual OptiX, forward/deferred comparisons and equivalent-output batch scaling demonstrate no GPU speed advantage for this scene and implementation. Native graphical learning takes 3066.60 s, including reference audits. The standalone Blender addon also recovers an interrupted owning process in Windows and a fresh Linux environment, reproducing 61 optimizer states and 150 modal powers exactly. We retain failed trials, restricted quadratic capacity and model limitations. AMD execution, universal native-transform and gradient error bounds, full-network physical fidelity, independent human review and exceptional novelty remain unestablished. A prospective confirmation with an independent mesh-free angular-spectrum solver, the same continuous device at every resolution and all 21 preregistered phases confirms that adding free paraxial diffraction to the path model predicts Mach-Zehnder modal powers within 0.0040 (budget 0.02), while the model without diffraction fails at 0.0281; the archived FDTD agrees within 0.0048. The same criteria place all 17,060 paths of the captured scene outside the diffraction-free regime at its represented wavelength. Lemmas T2-T5 behind the certificates are machine-checked in core Lean. The 16 trainable translations span a 12-dimensional decision family, against 26 for a free linear optical class, and lose 0.051 test accuracy to it on linear-teacher tasks; a compiled operator reproduces the traversal but is not faster when recompiled at every call.

1. Question and falsifiable scope

The operational question is whether an accessible scene editor can serve as a reproducible instrument for an own optical neural model, with traceable geometry, coherent fields, learning and useful numerical certificates. The represented model is ideal, scalar, monochromatic, lossless planar optics in a homogeneous medium. It is not a calibrated photonic processor or a complete Maxwell model.

The certificate claim is falsifiable: one independently verified exact represented-model output outside a claimed enclosure, one invalid certified decision, or one unbounded omitted nonzero admitted branch refutes that claim. Completeness, geometric correctness, enclosure correctness and classifier accuracy are separate acceptance conditions. A solver that rejects every useful input does not demonstrate utility. A timeout is not a zero optical field or a scientific success.

The original 90-second pilot was executed without extending its deadline and remains inconclusive. The exact indexed variant also timed out. Subsequent finite-state graph, training, GPU and wave protocols were distinct, frozen and published before execution under explicit human GitHub authorization. External registration/IPFS remains an additional pending route; no external identifier is claimed, and a later timestamp would not retrospectively preregister earlier observations.

2. Model, units and representation

2.1 Evaluated scene ingress

The capture reads Blender's evaluated dependency graph: represented mesh vertices, triangle indices, instance transforms, object identities, resolved input/parameter/detector bindings and scene unit metadata. Hexadecimal floating-point values preserve exposed numbers. Rational arithmetic constructs world geometry from those represented values. This defines an exact affine represented model; it does not certify the unobserved geometry intended by an author or reproduce every rounding operation in Blender's native transform/render pipeline.

The current scene admits 104 optical objects and 6,656 triangles, five sources, eight terminal modes and 32 coordinate bindings, grouped into 16 pairs of mirrors. Coordinates and wavelength use Blender units (BU); λ is the represented value of 0.1 BU. Scene display scale is not measured physical calibration. Reported powers are normalized scalar modal powers, not calibrated watts.

2.2 Coherence and detector semantics

Inputs are explicit complex amplitudes in one common launch-phase reference. The temporal convention is $\exp(-i\omega t)$; propagation contributes $\exp(+i2\pi L/\lambda)$. Declared mirrors and splitters use explicit phase/sign conventions. Summing intensities from independently labelled sources would remove cross terms and implement a different classifier. Fluence alone does not determine the complex field.

For a terminal mode, the field is the coherent sum of its admitted path contributions and power is its squared magnitude. Geometrical mode projection and completeness precede readout. The three class detectors are fixed. An estimated argmax is not automatically certified: certification requires the winning lower power bound to exceed every competing upper bound. Zero input and overlapping bounds retain an unknown decision.

2.3 Complete finite-state propagation

A state identifies represented origin, direction and previous optical plane. Two states merge only when these quantities are exactly identical, so their future propagation operator is identical. Every nonzero transmission/reflection remains an edge. Only mathematically exact zero coefficients omit an edge. Cycles, unresolved intersections, missing support or resource limits produce incomplete status with no fabricated field.

An exact conservative object-AABB index narrows candidate work without changing the closed triangle queries. Shared triangle vertices do not become artificial optical edges: an independent projected square-perimeter witness checks an interior neighbourhood of the union of coincident triangles. Nearest-hit and branch auditing uses a separate implementation. The captured complete graph has 133 states and 186 edges while preserving 17,060 terminal path occurrences. This is a representation of the admitted scalar network, not a proof that a real optical device has only those physical paths.

2.4 Geometry-driven derivatives and training

For fixed plane normals and admitted paired translations, ray/plane intersections yield exact affine origin, hit and segment-length derivatives. A merge is admitted only when its incoming affine origin derivatives agree. The phase derivative follows from the geometrically derived optical length. Own coherent forward derivatives propagate through every graph edge; power derivatives use $2\,\mathrm{Re}(E^* \partial E/\partial d)$. No historical analytic weight matrix supplies the trained field or its gradients.

Training minimizes cross entropy of the three detector powers divided by a fixed loss temperature. Adam updates 16 mirror-pair world-X translations. Proposed positions are quantized to Blender's represented native float32 coordinates and constrained around a declared untrained centre. Every evaluated optimizer state is independently audited; a fresh final geometric rebuild checks the affine computation. Finite differences and a separate functional autograd reference are software/numerical controls, not rigorous universal gradient certificates.

2.5 Capacity limitation

At fixed geometry, the network is linear in complex input: $E = Ux$. Detection introduces a quadratic readout. For real input, one detector has $P = x^{\mathsf{T}}Qx$ with $Q = aa^{\mathsf{T}} + bb^{\mathsf{T}}$ when its field row is $a + ib$. Consequently Q is positive semidefinite with rank at most two; a difference between two such powers has rank at most four. For

complex input, the corresponding Hermitian detector matrix has rank at most one. These algebraic restrictions hold independently of a numerical rank threshold.

The current model has no intermediate optical nonlinear activation and does not establish universal neural approximation. Normalizing the common input power is part of the declared encoding, shared with the classical comparisons. A dense matrix compiled from the graph's five basis responses is an equivalent fixed-geometry baseline, useful for auditing and cost comparison.

A separately frozen independent five-basis interval composition of the final native trained graph encloses the full 8×5 operator. Hermitian interval LDL certifies column rank five. Gershgorin bounds enclose every eigenvalue of U†U within $[1-\varepsilon, 1+\varepsilon]$, $\varepsilon <= 3.0665868333669135\text{e-}19$, proving $|\|Ux\|^2-\|x\|^2| <= \varepsilon\|x\|^2$ for every complex represented input x. This fixed-geometry approximate isometry includes all eight terminal modes; the three class detectors are only a subset. It is not an exact identity or calibrated physical-flux result. Midpoint eigensolver values are diagnostic only; a proposed vector is separately checked with rational Rayleigh intervals. Full analysis costs 6.468 s and 46.207 MiB RSS.

3. Independent numerical checks

The independent output enclosure uses rational outward rounding at 128 bits, interval π from Machin series with remainder, square-root bounds from integer arithmetic and exact rational Taylor evaluation with explicit remainders. Complex sums and powers are composed through the audited graph. This bounds an already observed native output against the exact represented model rather than claiming that every libm or GPU execution has the same error.

The first enclosure was valid but too loose: field L1 bound 1.51×10^{-7} and power bound 1.93×10^{-7} exceeded the fixed 10^{-11} budgets. It remains archived. The revised exact-midpoint Taylor composition was published before the secondary recomputation, with the budgets unchanged. The final maximum bounds were $4.9383434427813286\times10^{-15}$ for complex-field L1 error and $3.1372973883238074\times10^{-15}$ for power error. The represented unit-input decision R1 had a lower margin of 0.18933601504039158. A separately written 90-digit full-graph composition fell inside every reported interval. The rational construction supplies inclusion; the high-precision computation is a cross-check.

An independent trained-batch certificate covers all 150 observed fields, powers and ideal train-only encodings: maximum field L1 bound 2.7683×10^{-15} and power bound 1.0366×10^{-15}. All decisions separate; 137 are correct and 13 wrong. Its input reference begins with parsed binary64 features, rather than unobserved measurement values.

These certificates do not fill missing budgets for intended geometry, native transform preimages, uncertain physical parameters, polarization, diffraction, detector coupling or material calibration. Those errors remain unknown. The unit-input decision certificate is not a certificate of all Iris or Wine predictions.

A separately frozen conditional robustness study defines 16 paired-translation boxes by half-adjacent binary32 coordinate spacing, expanded by factors 1, 10, 100 and 1000. Independently derived exact affine plane expressions prove selected surface-union support and exclude all 13,699 competitor-object opportunities across 133 states per box. All four continuous topologies are proved; interval coherent fields preserve 147/150, 131/150, 40/150 and 0/150 class argmax decisions respectively. Remaining decisions are unknown, not demonstrated unstable. At the smallest box (maximum shift 9.537×10^{-7} BU), 137 certified decisions are correct and ten are wrong. These are conditional represented perturbations, not verified full Blender transform/mesh preimages or manufacturing tolerances. Conservative field/power variation bounds are distinct from observed arithmetic error budgets.

A separate whole-family certificate proves the original training box (centre ±0.035 BU plus native quantization allowance), unioned with all initial finite-difference offsets. All 133 state topologies and competitor exclusions are proved without shrinking the region after observing outcomes. Independently derived exact bases and all 16 coefficients agree for 1,251 scalar affine expressions. The full certificate costs 71.968 s and 49.160 MiB RSS. This justifies preparing a distinct reuse experiment with one continuous proof and per-state exact membership checks; The subsequent separately frozen full CPU pair preserves every coordinate and loss across 61 optimizer states, and all 150 final powers and predictions exactly. Complete worker costs are 393.110 s with the original point audits and 90.929 s with a freshly executed continuous proof plus 61 exact membership checks, an observed ratio of 4.3233. The fresh continuous proof and final reconstruction are included. This single shared-machine pair has

no timing confidence interval, energy measurement or universal-speed claim; the installed 0.1.2 add-on retains its original trainer.

4. Learning and generalization experiments

4.1 Iris geometry training and native roundtrip

The declared Iris split contains 120 training and 30 held-out examples. Training-only min/max preprocessing maps four features to four coherent sources plus a fixed coherent reference, with joint unit input power and no held-out clipping. Iris was previously examined in the project, so this experiment is not presented as a new blind discovery.

Historical learned delays were reset to an explicit untrained absolute geometry before initialization. Sixty fixed Adam updates decreased training loss from 6.2554405 to 0.3097067; training accuracy was 110/120 and held-out accuracy 27/30. All 61 represented states passed their geometry audits. The maximum initial loss-gradient finite-difference discrepancy was 1.4746713×10^{-6} against a 10^{-4} tolerance. Final fresh-geometry field reconstruction differed by at most 4.6194484×10^{-16} against 10^{-11}. Full worker cost was 369.43 s, including 349.67 s of geometric auditing.

Actual Blender 4.5.14 applied the final native coordinates, saved a new file, reopened it with script execution disabled, recaptured geometry and rebuilt the own inference graph. All 150 observed powers had maximum difference zero and all predictions matched. The original .blend remained unchanged. This proves a bounded local native roundtrip, not external replication. A separately frozen secondary analysis independently audits the final native graph and certifies all 150 recorded fields, powers and class argmax decisions. Ideal train-only feature encoding from parsed binary64 CSV values is also enclosed; observed encoder error is at most 1.579×10^{-16}, field L1 error 2.7683×10^{-15} and power error 1.0366×10^{-15} within unchanged budgets. The minimum decision margin is at least 2.11197×10^{-4}. Of the 150 certified decisions, 137 are correct and 13 wrong: numerical stability does not imply class correctness. Input measurement/parsing uncertainty and intended/native-transform geometry errors remain unknown.

4.2 Prospectively fixed Wine comparison

Wine contains 178 examples and 13 measurements. The official original files and hashes are retained with attribution and licensing. Four features were fixed before training: alcohol, malic acid, ash and alkalinity of ash. One stratified split fixed 141 training and 37 held-out examples, and three initializations fixed seeds 1049, 1050 and 1051. The same training-only encoding, sources, class labels and split were used for all models. No held-out result selected an initialization, feature subset, step count or hyperparameter.

Linear softmax used 15 nominal coefficients; quadratic softmax used the 15 unique input monomials and 45 nominal coefficients. Both optimized a declared convex L2-regularized objective using fixed L-BFGS-B controls. Their function families differ from a passive optical network, so matching data does not establish equal physical constraints or capacity.

Model	Training correct	Held-out correct	Held-out Wilson 95% interval
Geometric, seed 1049	117/141	33/37	75.3–95.7%
Geometric, seed 1050	108/141	30/37	65.8–90.5%
Geometric, seed 1051	105/141	28/37	59.9–86.6%
Linear softmax	120/141	32/37	72.0–94.1%
Quadratic softmax	121/141	32/37	72.0–94.1%
Training majority	56/141	15/37	26.3–56.5%

All 183 geometry states and three final rebuilds passed. Maximum initial gradient discrepancy was 5.473×10^{-6}; fresh field difference was at most 5.882×10^{-16}. The complete worker cost was 1114.17 s, including 1054.49 s of geometric audits. Baseline fits took 5.58 ms and 4.71 ms after imports, with imports included only in the total worker cost.

The three runs share the same 37 held-out observations and cannot be pooled as 111 independent trials. Paired exploratory exact McNemar p-values against either baseline were 1, 0.5 and 0.125, without multiple-comparison correction. No optical superiority is demonstrated. The spread across initializations shows sensitivity, and the single public-data split does not establish external blind or zero-shot generalization.

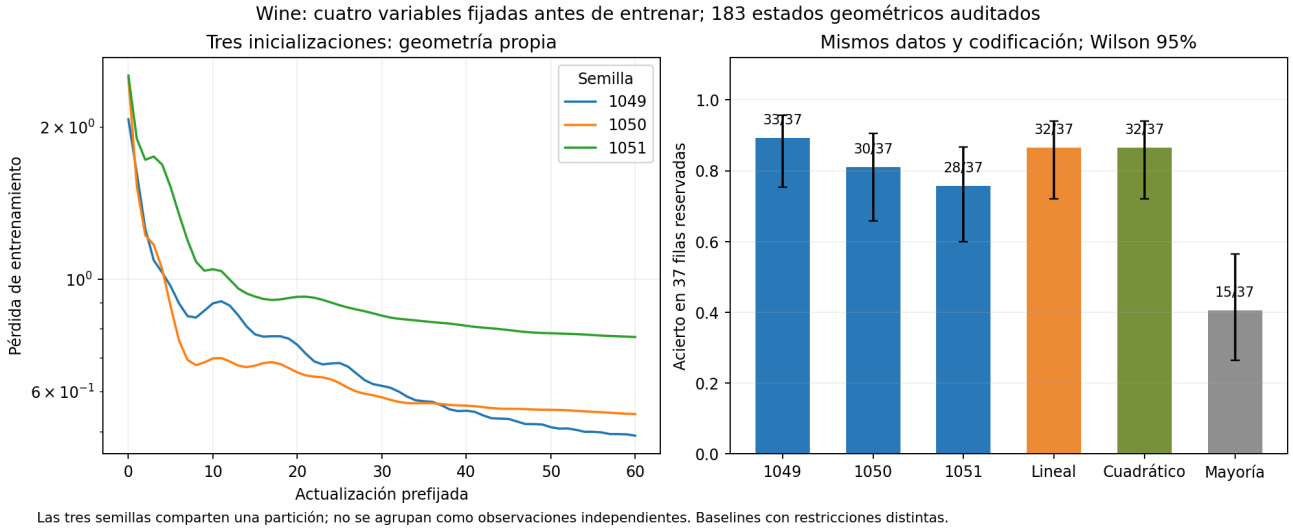


Figure 1: All three Wine runs and matched comparisons

5. Actual NVIDIA execution and total-cost limits

The actual RTX 3090 was fixed by device UUID and admitted through the shared FIFO queue. Torch 2.6.0+cu124 with CUDA 12.4 and driver 581.29 evaluated complex128 fields, powers and the own 16-parameter Jacobians for 60 prepublished complex probes. Exact geometry and phase-argument preparation remained on CPU; this was not GPU triangle tracing. A separate functional autograd reference checked the weighted-loss derivative.

Maximum CUDA/NumPy differences were 3.61×10^{-16} in field, 3.89×10^{-16} in power, 5.55×10^{-14} in field Jacobian and 3.55×10^{-14} in power Jacobian; the autograd discrepancy was 1.56×10^{-13}. Independent rational certification of these 60 observed GPU probes gave field L1 bound $\leq2.9356883\times10^{-15}$ and power bound $\leq1.3807709\times10^{-15}$. Fifty-nine represented argmax decisions were certified; zero input remained unknown.

Batch	Cached graph CPU, ms	Cached graph CUDA, ms	Equivalent dense CPU, ms	Equivalent dense CUDA, ms
1	14.45	24.50	0.071	0.435
150	17.43	27.06	0.499	0.559
2048	75.00	25.72	8.597	0.897

These are synchronized warm medians of five fixed repetitions for the same input/output contracts, including fields, powers and Jacobians. Preparation, transfers and complete worker costs are retained separately. The whole GPU benchmark worker cost was 6.72 s; peak RSS was 996.10 MiB and peak reserved CUDA memory 112 MiB.

Small graph batches are slower on CUDA, and the dense baseline is more efficient at all reported sizes. Warm batch scaling is not structural graph scaling, full native geometry-training speedup, host/device energy measurement or photonic-device efficiency. AMD hardware was not available and has not been executed.

A separately frozen full training pair executes the unchanged CPU trainer and own CUDA manual cross-entropy/Jacobian/Adam arithmetic for 60 updates, with 61 exact CPU geometry audits each. Both achieve 27/30 held-out predictions and identical final native-coordinate deltas; all 150 decisions agree, maximum power difference 3.8858×10^{-16}. Across CUDA states, maximum matched CPU field/power/loss/gradient differences

are 5.404×10^{-16},6.662×10^{-16},8.882×10^{-16} and 2.843×10^{-13}. Fresh final geometry agrees within 5.118×10^{-16}. Full-worker CPU/CUDA times are 370.46/390.09 s; this pair provides no global speedup. Geometry audits dominate at 350.49/363.40 s. CUDA loss/gradient/proposal time is 2.12 s, versus 1.17 s for the CPU training updates, with phase preparation/transfers and CPU controls also retained. GPU reserved memory peaks at 22 MiB. This is one pair, not a timing confidence interval; exact geometry and native quantization remain CPU.

A subsequently frozen native graphics frontier constructs the network directly from the five source rays, with new reflection/transmission queries rendered against the 6,656 evaluated triangles in 17 batches. Three actual RTX3090 native OpenGL constructions each exactly verify all 133 graphics surface/primitive candidates without CPU replacement, producing 133 states, 186 edges and 17,060 represented terminal paths. Exact nodes match fresh CPU and prior native geometry; an independent full graph audit passes. All 150 phase-aware outputs preserve predictions, with maximum field/power differences 5.1179e-16/5.5512e-16. Coherent phase arithmetic uses CPU exact refined lengths rather than graphical depth. Full fresh CPU construction costs are 30.7660/30.4246/29.8658 s versus 31.5350/32.1434/33.0084 s for GPU graphics plus exact verification: no full construction speedup is observed. Candidate verification dominates the latter at 30.5500/31.1753/32.0184 s. This is actual graphical geometry execution, not matrix multiplication substituted for a scene, but it does not imply hardware RT cores, physical fidelity or novel graphics tracing. A separate frozen close-plane diagnostic selected the wrong plane at a 2ˆ-25 BU spacing; that failure is retained and motivates exact verification.

A separately frozen optimizer recovery trial actually terminates only an owned research worker after its atomic checkpoint commits 12 updates. A fresh process repeats the whole original-family proof and deterministically recomputes the 12-update prefix before accepting coordinates and Adam moments. All 61 coordinate/loss states and all 150 final powers/predictions equal the uninterrupted reference exactly; both obtain 27/30 held-out correct. Four actual checkpoint adversaries are rejected, including forged optimizer moments with a recomputed self-consistent hash. Original checkpoint bytes are preserved. Full uninterrupted/interrupted/resumed costs are 96.8763/75.5020/94.5633 s, total 267.2220 s and peak owned RSS 56.59 MiB. This validates process-level research-worker recovery, not yet the installed Blender resume UI, whole-host crashes or power-loss durability. The first frozen attempt failed in final metadata before the interruption stage; that null outcome is retained.

A prospective second native graphics profile replaces the expensive candidate fan-support routine with exact plane intersection, independent Gram closed-triangle predicates and a positive square-perimeter interior witness, followed by unchanged exhaustive independent nearest-object exclusion. Three original CPU builds cost 29.8096/30.0367/29.9906 s versus 10.5964/10.5326/10.4500 s for GPU graphics plus the new exact refinement, with all gates, 133 states, 17,060 paths and 150 outputs preserved. This approximately 2.85 median-ratio observation is a combined algorithmic-route improvement, not an isolated GPU benefit: that observation alone leaves the identical-refinement candidate-engine ablation unresolved; the subsequent study below tests it. Three shared-host repetitions with fixed ordering do not establish a general speed factor. Full process cost is 134.7320 s and peak owned RSS 348.40 MiB.

A subsequent prospectively fixed alternating three-pair ablation uses native Blender CPU BVHs and GPU raster depth candidates with identical exact selected-plane, Gram, interior and exhaustive-nearest refinement. All six graphs and all 150 outputs pass. Full CPU construction costs 9.897551/9.829980/9.868582 s versus 9.964846/10.004324/9.912861 s for GPU construction (median CPU/GPU ratio 0.99034). No global GPU advantage is observed in this scene. Proof dominates both routes at approximately 9 s. The full supervised process costs 72.844940 s with peak owned RSS 364.90 MiB. This controls the earlier combined improvement and precludes attributing its 2.85 ratio to GPU hardware. Three alternating exploratory pairs do not establish universal scaling or energy benefits.

A further separately frozen trial executes coherent transport in actual native OpenGL fragment shaders. The shader receives captured planes/materials, current geometric rays, wavelength and incoming complex fields. Raster/depth selects each surface; FP64 arithmetic derives optical distance, phase, splitting, mirror phase and terminal-reference phase. No expected hit IDs, precomputed transfer matrix or edge phase weights supply that computation. Exact CPU topology admission and compensated coherent merging remain explicit. Three 150-input repetitions each make 19,950 actual geometric queries and preserve every native prediction, with maximum complex-field discrepancy 7.5011e-14 and power discrepancy 7.2165e-15, below unchanged 1e-11 gates. Four actual GPU phase/interference controls pass; coherent versus intensity-sum power differs by up to 0.27376. Transport plus CPU merging costs 2.5075/3.2332/3.4803 s; full supervised cost is 36.6185 s and peak owned RSS 403.04 MiB. CPU phase-only graph reuse is a different workload and supplies no equivalent speedup claim. Ten preceding null

trials preserve interface, integer-readback and shader-lifecycle failures. The successful trial verifies all binary64 input echoes and rejects every newly raised GL error. GPU gradients, rigorous bounds on unobserved GPU rounding, physical fidelity, AMD and superiority remain unestablished. Protocol and retained evidence.

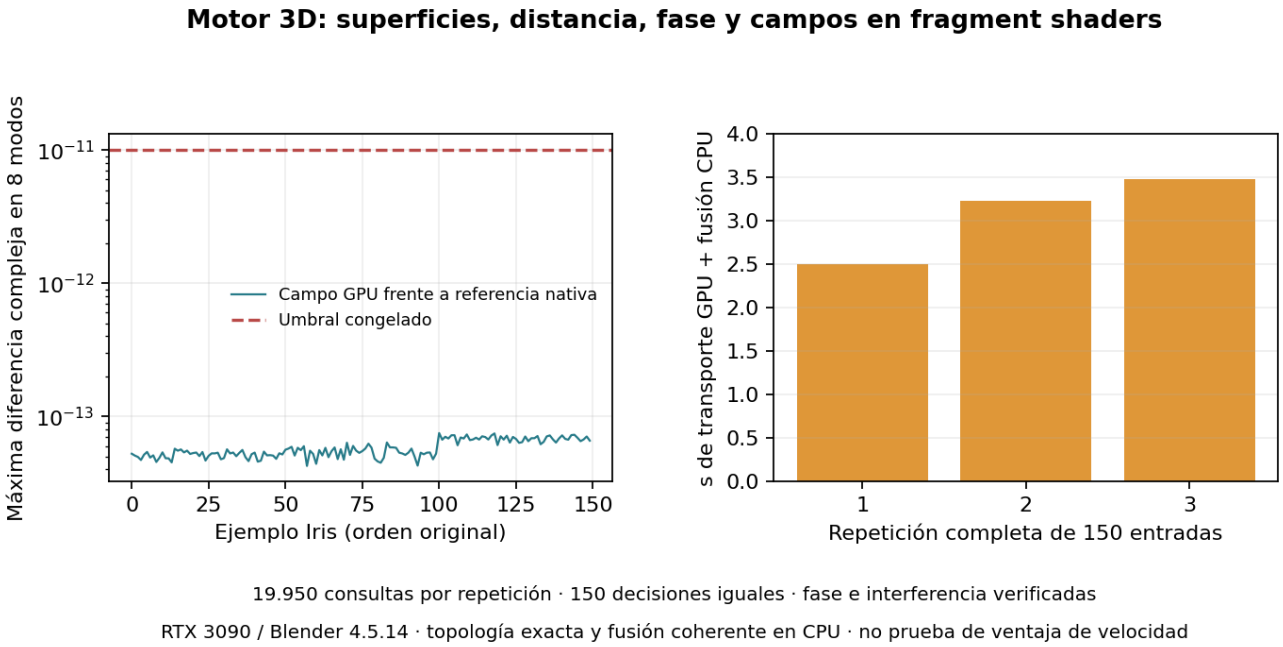


Figure 2: Captured-surface coherent fragment transport

A targeted additional primary-source review establishes relevant graphics/optical precedents: Purcell et al. 2002 maps ray tracing to programmable graphics stages; MacroSim institutional documentation describes double-precision GPU/OptiX optical tracing and a 3D scene editor; Mauch et al. 2014 explicitly includes coherent tracing in its publisher abstract; Keksel et al. 2023 models optical instruments with configurable OptiX laws and compares virtual images with its physical focus-variation instrument. These precedents rule out treating graphics-engine optics or coherent GPU tracing alone as our original contribution. No implementation was integrated or reproduced. The full 2014 article was inaccessible; the reviewed 2023 method passages do not establish equivalence to this network. Primary Crossref metadata confirms the 2013 MacroSim DOI 10.1117/1.OE.52.5.053004, correcting the mismatched institutional link. Review and source-access receipt. The historical literature extraction backlog remains unchanged and novelty unestablished.

A separately fixed independent rational secondary analysis audits the native geometry and bounds all 450 observed graphical outputs, after exact equality of the three repeated field/power arrays is checked. Maximum field L1 error is bounded by 9.9736e-14 for recorded binary64 inputs and 9.9745e-14 for ideal train-only encoding; corresponding power bounds are 7.1507e-15 and 7.0918e-15. All four phase/coherence controls pass their interval error budgets. All 150 unique argmax decisions are certified, including 137 correct and 13 wrong predictions; repeated runs do not add independent biological observations. The minimum certified margin is at least 0.00021119712. Full analysis costs 21.0174 s with peak owned RSS 105.49 MiB. These certificates enclose the observed GPU arithmetic and CPU merges against the exact represented model, not every unobserved GPU execution or physical optics. Full interval evidence.

Native captured-surface graphical derivatives

A separately frozen native graphics trial differentiates all 16 paired-mirror X translations for five fixed probes (three native Iris rows, one bounded complex input and zero). FP64 fragment shaders derive intersection-distance, phase and complex-field tangents from captured planes/materials and current incoming value/tangent. Exact CPU affine origin jets, topology admission, compensated coherent merges and the power/loss chain rule are explicit. No transfer or optical Jacobian matrix is fed to the shader. All gates pass: field discrepancy 6.5773e-14, field-Jacobian discrepancy 5.7353e-12, power-Jacobian discrepancy 9.2771e-13 and diagnostic loss-gradient discrepancy

5.4321e-12. Thirty-two independently audited materialized perturbed geometries give central finite-difference scaled discrepancies 7.4685e-10/1.2831e-9/2.6588e-9 for field/power/loss. Zero-input derivatives are exactly zero. The 16 passes execute 21,280 geometric queries; full supervised cost 336.9243 s and peak owned RSS 434.86 MiB include 296.7458 s of independent finite-difference audits. All 8,084 owned GL stages and exact input echoes pass. This validates observed hybrid graphical derivatives for the stated probes; it does not demonstrate full native graphics training or universally enclose derivatives. Protocol and raw evidence.

Complete native graphical learning and actual acceleration components

A distinct, prospectively published native deferred graphics protocol retains the original 120/30 split, train-only encoding, absolute untrained initialization, temperature 0.05, 60 updates and quantized paired world-X parameter bounds. Each state from 0 to 60 is applied to the actual Blender scene and recaptured. All optical objects match exact affine materialization and belong to the independently proved original 16-dimensional topology family. Actual GPU candidate caches reset for each geometry generation, including the final save/reopen: 62 generations. Their keys contain exact represented rays and previous-plane exclusion; every cache fill retains its native raster readback. A fullscreen FP64 stage derives distance, phase and optical tangents from these actual selections and current planes and materials. No CPU expected hit, phase, optical transfer matrix or optical Jacobian enters the shader. This uses the established separation of geometry and shading in deferred shading; that technique is not our novelty.

All 60 updates use the graphical field and power tangents. Across all 61 states, maximum differences from the independent CPU numerical reference are 8.99309e-14 for field, 7.60358e-12 for field Jacobian, 1.29674e-12 for power Jacobian and 1.03455e-11 for loss gradient. The GPU-driven quantized positions match the fixed CPU reference optimizer exactly. Loss decreases from 6.25544053835 to 0.309706672112; training accuracy is 110/120 and held-out accuracy is 27/30. Held-out labels are scored only after the final update. All 150 final decisions match both the independently rebuilt saved/reopened CPU graph and the reference optimizer. These derivatives describe a continuous world-X extension at the admitted represented positions, rather than a derivative of Blender's discrete quantizer. All 1,030,023 owned GL checks and input echoes pass. Numerical parity is an observation, not a universal gradient enclosure.

The complete worker-launch-to-exit supervision takes 3066.598253 s, with peak owned RSS 499.45 MiB. The 61 graphical transports and compensated CPU merges take 2692.497419 s; CPU numerical references take 3.093286 s, and native captures take 113.688903 s. Setup, final graph reconstruction, independent audits and readback remain separately recorded. There are 8,246 actual triangle surface queries and 31,193,820 optical fragments including echoes; these counters measure different operations. Queue wait, preflight, source copying and publication are outside the timed optical inference arms. Protocol, all states and raw hashes.

The first full graphical attempt was stopped after its first state took 322.45 s and projected beyond the frozen 7,200 s envelope; its complete-trial metric remains null. A later comparison on identical geometry, 120 inputs and 16 parameters returned bit-identical fields, Jacobians and loss gradients between deferred and forward arms. Deferred took 42.973787 s, compared with 38.272944 s for forward. The historical 322 s observation therefore cannot establish that deferred shading accelerated learning. We retain both the failed trial and this negative comparison. Separately, actual Cycles OptiX on RTX 3090 reconstructs all 133 states from captured-triangle renders followed by exact admission. All 150 decisions match; candidates plus identical proof cost 41.008216 s, compared with 10.330787 s for CPU BVH. This OptiX component computes coherent fields in CPU and does not establish complete optical acceleration with hardware ray tracing. OptiX and its retained extent failure, equivalent forward/deferred comparison.

A separately frozen independent 128-bit outward-rounded rational analysis of the final trained native scene passes the unchanged input budget of 1e-12 and field/power budgets of 1e-11. For recorded input amplitudes, maximum field L1 error is 9.97351e-14 and power error is 7.15061e-15. Against the ideal train-only encoder, the respective bounds are 9.97449e-14 and 7.09172e-15. All 150 represented decisions separate: 137 botanical labels are correct and 13 are wrong, with 0 unknown decisions. The minimum lower margin is 0.000211197. This certifies observed final shader-plus-CPU-merge arithmetic, without certifying every training gradient, all unobserved GPU operations, quantizer preimages or intended physical geometry. Independent final certificate.

The equivalent-output scaling experiment uses cyclic repetitions of the previously observed Iris rows with a common exact quarter-turn phase. These are throughput inputs, rather than new independent examples. Both

arms return all eight complex fields, powers and identical decisions. CPU computes the geometry DAG and its current geometry-derived phases, without a dense precomputed transfer matrix. GPU retains the validated FP64 zero-tangent and bit-echo kernel, 150-row tiles and CPU coherent merging; that implementation overhead remains in its timed arm. Two fixed alternating orders per size are descriptive cold/warm observations, rather than statistical evidence for general performance. Full capture, fresh GPU graph, independent admission and setup costs are separately recorded.

Batch	CPU first / second (s)	GPU candidate-cache cold / warm (s)
1	0.026788 / 0.020904	0.592882 / 0.335261
150	0.020160 / 0.031941	4.027876 / 3.458444
4096	0.024342 / 0.026418	98.006908 / 97.720299

Maximum field difference is 7.50105e-14 and power difference is 7.21645e-15; every decision agrees. The current GPU implementation is slower in every measured pair. The initial RAM-floor interruption remains null; the accepted repeat used exactly the same frozen method and limits. Whole-GPU power samples include display and other activity, and do not isolate kernel energy or photonic-device efficiency. Fixed protocol, original failure and complete repeated run.

Secondary packet audit and rejected resident variant

A separately frozen CPU-only replay of archived actual fields and matching rays reproduced all 5,650 optical packet extents. The NumPy candidate and original packet builder produced identical bytes, including signed-zero, subnormal and extreme-finite controls. For 4096 inputs, packet generation was1.119/1.277s for the original and0.457/0.512s for the candidate; the candidate was slower for one input. These two alternating-order observations exclude input construction and graphics operations and do not establish an overall GPU speedup. Actual stage receipts reconcile5,834chunks and source-derived35,004integer reads/46,488texture creations, with191,168,512logical upload and144,652,928logical readback bytes. These are API payload extents, not measured PCIe bandwidth or causal runtime fractions. Protocol and raw evidence.

A new native variant derives local complex coefficients from actual GPU-selected surfaces, uploads those local coefficients with admitted topology, and executes all133optical states and Kahan coherent merges per input in a fragment shader; no global transfer matrix is supplied. Three prospectively frozen native attempts remain rejected. The first exceeded Blender's six-attachment framebuffer limit. Separate split-output versions executed two four-mode output passes and a five-channel exact input echo, but the coherence-control field error was6.376147888196662e-8 against an unchanged1e-11budget. All64diagnostic real/imaginary outputs lie on the binary32grid, while242of532local coefficient scalars do not; the cause of the precision loss is unresolved. Actual coefficients/controlfields and exact input echoes are archived before rejection. No complete inference, cost advantage, gradient validation or release is accepted for this variant. The earlier validated hybrid graphical engine remains the reference. All attempts and witnesses.

6. Separate wave-reference experiment

An original coherent complex128 angular-spectrum solver propagates a declared Gaussian boundary field in a homogeneous outgoing scalar Helmholtz half-space. Its continuum reference is independently implemented Fourier–Bessel/Hankel quadrature with analytic Gaussian spectrum and separate quadrature refinement. A paraxial Gaussian solution is an additional approximation reference. The transverse integral of $|E|^2$ in a circular aperture is the declared observable; it is not a vector Poynting flux or a captured detector's modal-coupling measurement.

The fixed family uses waist 0.1 BU, aperture radius 0.15 BU, wavelengths 0.1/0.025/0.00625 BU and propagation distances 1/2/4 BU. The first 36 observations completed, but the λ=0.1 BU, z=4 BU case failed the fixed 2×10^{-6} radial-field tolerance at window 16 BU: difference 2.2311874×10^{-6}. Refining $512^2/1024^2/2048^2$ at that window left the same error. This negative result was published before any remedial trial.

A separately frozen adaptive family used windows 32/64 BU and grids $1024^2/2048^2/4096^2$, with unchanged scientific tolerances and a larger declared RAM envelope. All nine physical cases passed their prefixed gates.

Maximum radial-field differences were 4.0866539×10^{-7} at window 32 and 9.2920364×10^{-8} at window 64, in the 65 checked points. Aperture integration remained sensitive to spatial resolution; the coarse $1024^2/32$ mesh had maximum fraction error 0.00289647, retained rather than called precise. Fine and expanded meshes satisfied their own gates. Complete remedial cost was 133.05 s, peak RSS 1866.79 MiB.

Quadrature refinement is numerical convergence evidence, not a rigorous universal FFT error bound. This Gaussian mode was not inferred from the network capture. The separate study therefore characterizes a declared propagation regime without establishing full-network physical fidelity or providing a correction factor for its detector powers.

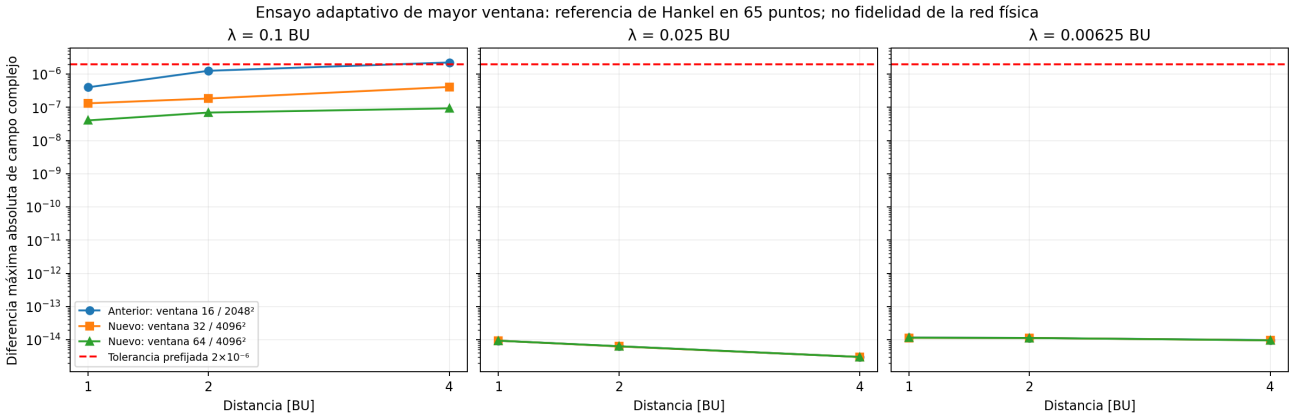


Figure 3: Retained negative and separately frozen remedial wave family

7. Installation, replication and artefact status

The standalone own Blender addon packages the original solver dependency closure, semantic contracts, own trained example, dataset, source provenance and integrity manifest. Its UI uses explicit coherent inputs, background Blender jobs, owned cancellation, stale-result rejection, completed-result recovery and new-file saving. Original implementations and immutable historical ZIPs are retained.

Two native attempts are retained with null metrics: a version-string/preflight failure before installation, and a successful isolated installation followed by an incorrect complex serialization handoff. ZIP 0.1.1 corrects that handoff with the already used coherent wire converter. Its third native audit passed all nine predefined controls in 729.01 s with peak aggregate RSS 573.49 MiB. Own training actually ran inside background Blender, audited all 61 states and reproduced the previous 27/30 held-out predictions, loss trajectory and all 150 powers with observed difference zero. Stale inputs/geometry, real cancellation, completed-job recovery, atomic apply, copy/reopen and modified-result rejection also passed. A subsequent shutdown log contains a double-unregister RuntimeError after the audit explicitly unregistered a still-enabled addon. It does not change the nine numerical/control outcomes, and it motivated a separately frozen lifecycle regression. ZIP 0.1.2 changes only lifecycle code, installation text and manifest; all scientific-worker, vendor and data bytes remain identical to 0.1.1. Five actual native lifecycle controls pass in 65.04 s with peak aggregate RSS 516.71 MiB, including duplicate registration, Blender utility cycles, owned-process cancellation on disable and native150 inference parity. The entire post-exit log contains no traceback or unregister error. The original failure remains archived. Bounded owner-process interruption recovery is tested below; human usability and electrical power-loss durability remain open.

Every bounded scientific record includes a published profile, source hashes, raw outcomes and evidence index. Reproducing old protocols requires their corresponding commit or archived preimages; current source evolution must not be substituted silently. A clean local installation is distinct from an execution in an independent environment, and both are distinct from a replication performed and interpreted by independent researchers. A separately frozen Linux CPU run on GitHub, published at commit `bbc06c3ace138d98873df9ba0a4d83252ebfefc8`, independently checks 40 source/profile/receipt blobs before science. All 34 selected controls, certificate replay and 61 training-state audits pass. It reproduces 27/30 held-out predictions and all 150 decisions, with maximum power difference 3.8858×10^{-16}; scientific-worker time is 367.37 s and peak aggregate RSS 88.29 MiB. Setup/upload are separate job stages. This is external-environment reproduction of our code, not independent human replication or

GPU execution. The original YAML setup failure is retained. A separate native Linux protocol published at `c1642ca090e8ba40bcd64f875b0d9ec39b4208ae` downloads and verifies the official Blender archive and executable before isolated ZIP 0.1.2 installation. All nine native controls, 61 training-state audits and the post-exit lifecycle log pass in actual Blender 4.5.14 LTS on Linux x86_64, Python 3.11.15 and NumPy 1.26.4. It achieves 27/30 held-out predictions; power differences are 3.8858×10^{-16} versus Windows and zero versus its final native geometry. Full controller cost is 685.77 s (21.09 s preparation, 663.69 s native controls), peak aggregate RSS 756.34 MiB. The first HTTP403 preparation failure is retained. This is external native software reproduction, not independent expert interpretation. The latter human replication has not been obtained. No specialist critique, journal submission, external registration receipt or physical experiment is asserted.

The standalone ZIP 0.1.3 adds an owned-worker guard and a real Resume operator. In a separately frozen Windows trial, the owning Blender process was terminated after checkpoint 12. Its worker independently exited with code 9 while the parent's process handle remained open. A fresh installed Blender re-proved the original geometry family, exactly replayed the prefix, rejected stale, truncated and forged histories, restored Adam, and applied, saved and reopened the trained scene. All 61 optimizer coordinates and loss values, compared by float-hex representation, and all 150 modal powers match the uninterrupted baseline exactly; held-out accuracy remains 27/30. Full supervision takes 695.905176 s and peak owned RSS is 574.68 MiB. A creation-time-only owner predicate failed a held-handle control; that negative and the corrected process-liveness check are retained.

The unchanged 0.1.3 archive and its SHA256 manifest also pass a separately frozen fresh Linux GitHub run, 37919893102. All 49 source, profile and receipt Git blobs are checked before execution. A process-local subreaper observes only its own orphan worker's actual exit with code 9. The newly installed Resume operator again reproduces 61 optimizer states and 150 powers exactly against the independent Linux uninterrupted baseline, with 27/30 held out. Stale, corrupt, forged, apply, save, reopen and post-exit log controls all pass. Full supervision takes 652.831612 s, including 20.980969 s of official preparation and 630.903356 s of native control; peak aggregate owned RSS is 787.05 MiB. The original artifact digest and all outer and native raw hashes are permanently archived. This is automated external-environment software reproduction, rather than independent human interpretation or graphical GPU/AMD reproduction. The installed 0.1.3 UI runs on CPU; the graphical research route is a separately validated command-line executor. Repeated interruption of an already resumed job and electrical power-loss durability remain open. Windows recovery, fresh Linux recovery.

8. Related work and novelty boundary

Trainable coherent photonic networks and diffractive optical learning predate this implementation. Shen et al. and Lin et al. establish that context. Kawata and Hirose is an earlier coherent path-difference learning precedent. These works prevent presenting optical learning itself as our novelty.

Ho et al. combines ray and diffraction-based optical modelling with optimization and coherent demonstrations, an important closest-method comparison. Its image-formation conventions and sampling scope cannot simply replace the common-coherence multi-input readout here. Chromatix provides a separate differentiable wave-optics precedent. We have not installed or benchmarked it in this study.

DeepG and QA-IBP show related certification precedents for geometric transformations and quantized inference. Their reviewed scopes do not establish exact optical triangle-path equivalence, but they prevent broad claims that combining geometry, decisions or actual arithmetic certification is automatically unprecedented. Matsushima and Shimobaba is a known angular-spectrum sampling-method reference; this work does not claim to have implemented its band-limited method or completed a full-text review of that paper. Steck's optics notes support the declared Gaussian conventions.

Knoll et al. establishes interval/affine inclusion for ray tracing. de Koning et al. optimizes optical geometry with differentiable non-sequential rays and reports LightTools checks. Chen et al., RFDT is a close coherent multipath/geometry-gradient precedent with edge-diffraction transitions. Those transitions differ from our hard-branch admissibility boxes; equivalence is not assumed. A bounded primary review records inspected passages and file hashes. These three supplementary identities have not been merged or deduplicated into the historical counts below.

The bounded literature register contains 194 raw records and 175 unique identities after deduplication. Thirty selected extraction records comprise 29 papers or discussions and one book chapter; 16 selections belong to the search cohort and 14 arose from directed searches. Another 106 cohort records are eligible background without a

full method extraction. Relevant primary method, result and limitation passages were inspected; the retrieved full texts were not all read in their entirety. These counts describe the historical audit, rather than an exhaustive absence-of-prior-art proof. The implemented combination is original code and an integration contribution. Exceptional scientific originality, exceptional importance and durable impact have not been demonstrated. A defensible novelty claim requires closer matched comparisons and independent expert assessment.

9. Conclusions and open tests

The bounded evidence supports an own geometry-driven coherent scalar classifier, exact admitted branch coverage, useful certificates of observed represented outputs, own gradients/training, local native Blender roundtrip and actual NVIDIA evaluation. It also exposes meaningful limitations: optical comparisons do not show superiority, small CUDA batches are slower, finite FFT windows can fail, and the current passive readout has restricted quadratic capacity.

The remaining research programme includes bounded input/parameter uncertainty and native-transform error budgets, bounded trained-batch uncertainty beyond the completed arithmetic certificate, broader geometry/task families, broader geometry/dataset scaling and ablative replication beyond the completed native batch/engine comparisons, isolated full-cost energy measurements, actual AMD execution, GPU UI integration and broader standalone/human acceptance, independent-environment graphicalGPU reproduction beyond the completed CPU native Linux host-recovery trials, outside expert replication, full-network physical modelling and calibrated measurements when physical-device claims are sought. None is replaced by a prize aspiration or by a passing software test.

Data, code and reproduction map

Repository: Agnuxo1/Neuro3D, main with scientific history preserved; PR history 6. Software license: MIT; Wine data attribution/license: UCI Wine, DOI 10.24432/C5PC7J. No third-party optical implementation is bundled in the own addon.

Evidence	Protocol / method	Raw record
Represented output intervals	method and limitations	analysis02 index
Observed native batch and ideal encoder	150-row certificate	certificate
Conditional continuous geometry boxes	fixed protocol	four boxes, unknowns retained
Own Iris training	frozen protocol	training result
Actual native save/reopen	native reproduction	native result
Wine and matched baselines	comparison	complete result
Actual CUDA training and paired full cost	training protocol	CPU/CUDA pair
Complete operator rank and normalization	Gram protocol	rational operator certificate
Whole continuous training family	family protocol	exact affine/topology proof
Fresh continuous-proof training full cost	paired protocol	full equivalent pair
Identical-proof native CPU/GPU geometry	prospective ablation	six full builds
Captured-surface coherent fragment fields	protocol and ten null trials	actual successful GPU trial
Observed graphics output and decision intervals	independent certificate	rational enclosure
Captured-surface graphics derivatives	frozen probe protocol	gradients and32 geometric finite differences
Actual NVIDIA	benchmark	indexed evidence
Gaussian negative/remedial	original, remedial	original result, remedial result

Evidence	Protocol / method	Raw record
Standalone addon acceptance	versioned protocol, failures and lifecycle limitation	nine-control native result
External native Linux Blender	fixed version, download failure and completed run	native external result
External Linux CPU reproduction	profile, failed setup and completed run	actual external result

Complete native graphical learning | prospective61actualstates | fullresult and allraw |
Final learned nativeGPU output intervals | independent final certificate | all150 certificates |
Native equal-output batch scaling | fixed1/150/4096 protocol | raw timing and fields |
Actual OptiX component and CPU cost | complete fixed trial | originalraw |
Installed0.1.3 owned-host recovery | Windows interruption/resume | exact states/powers |
External native0.1.3 owned-host recovery | fresh Linux protocol | originalZIP and rawhashes |

Publication commits fixed before experiments are recorded in each linked protocol. Hashes and preimages must be checked before interpreting or rerunning a result. External expert review and author approval of a submission-ready manuscript remain pending.

10. Dated consolidation and new CPU comparator evidence

The published two-dimensional P1-7 MZI refutes the scalar path model without diffraction in its tested regime: maximum absolute modal-power errors 0.0269264 and 0.0275190 exceed the unchanged 0.02 budget. The 12/18 discrete phases are not 21 exact equispaced physical geometries. Mesh-specific recalibration and the fixed-index reflectivity mismatch remain limitations. The exploratory Gaussian correction gives 0.0051349/0.0046153, requiring a new prospective device-level confirmation.

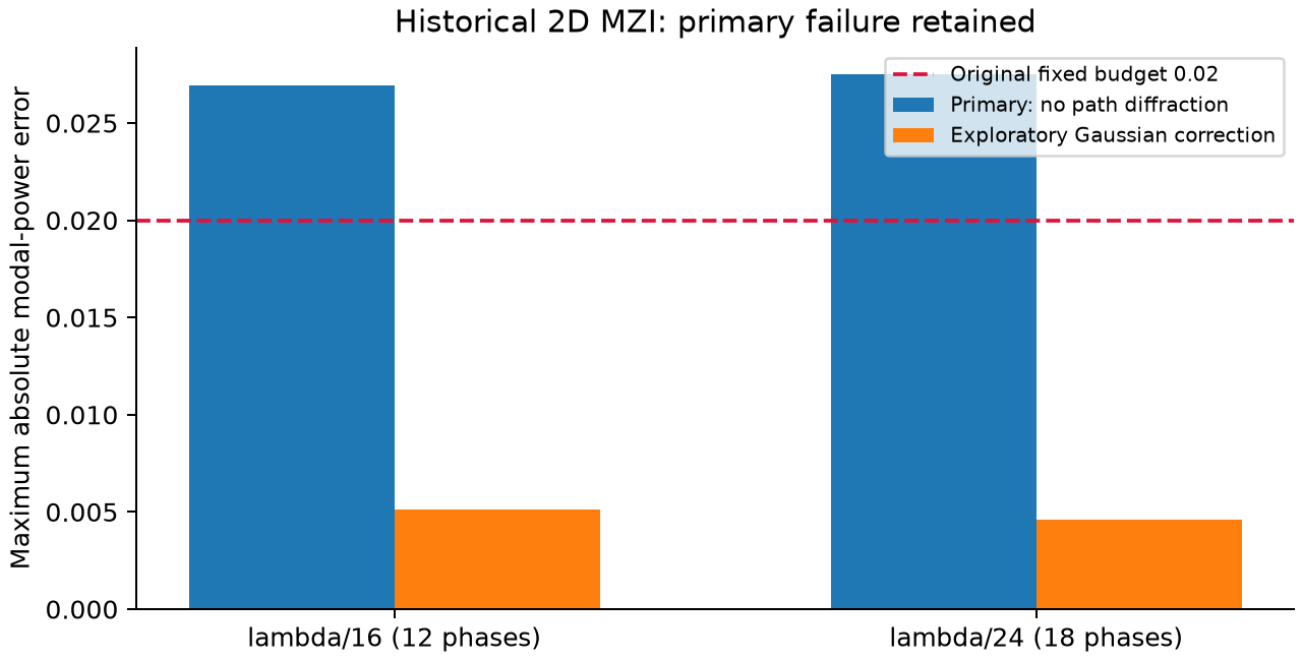


Figure 4: Historical wave negative and exploratory correction; this is not full-network validation

All six previously accepted same-output GPU/CPU inference pairs remain slower on GPU. They do not establish isolated energy or a photonic device advantage.

A prospectively published CPU ablation replays actual archived GPU coefficients through all 133 states for four coherent controls. The replay matches the archived CPU reference within 9.62378307390625e-14, below the

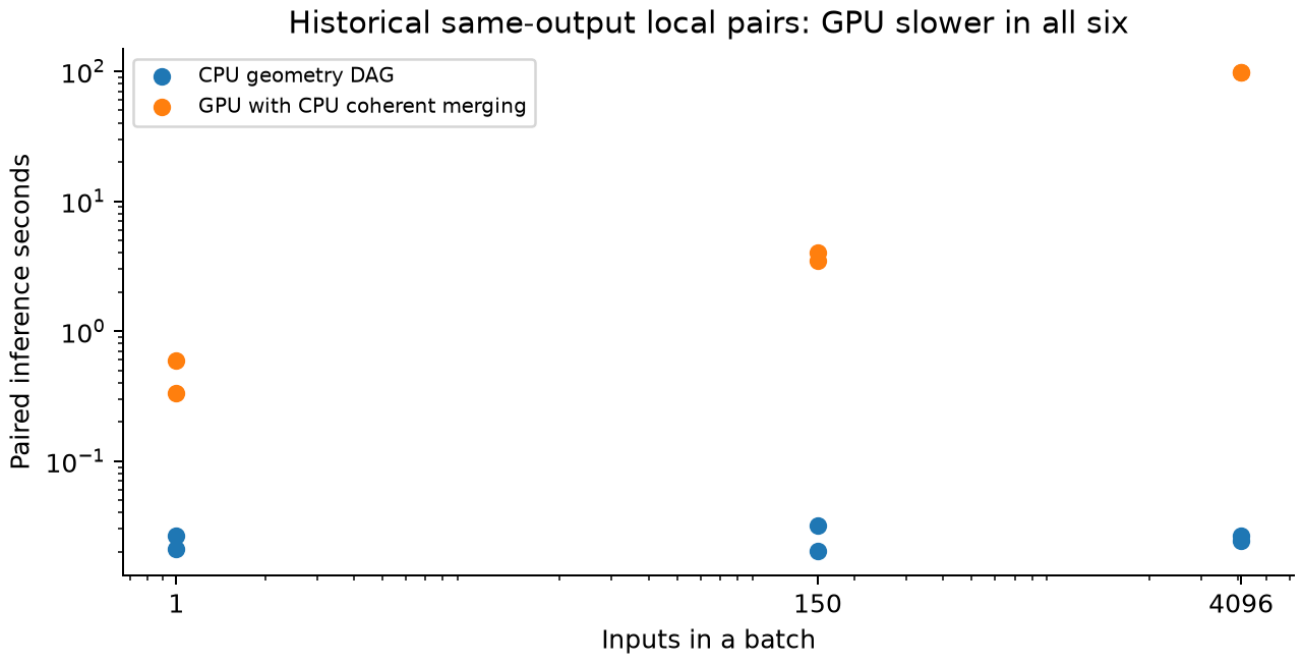


Figure 5: Historical inference costs; setup and proof costs are recorded separately in the protocol

fixed 1e-11 budget. It differs from archived native GPU outputs by 6.376146498978708e-8. Rounding only the final replay to binary32 does not exactly explain the observed outputs. Native intermediate-stage traces remain necessary; the resident backend is still rejected and the cause is UNKNOWN.

The newly completed preregistered FBCSP comparison uses all 17 existing subjects and 50 external LORO folds, the exact stored optical indices, training-only CSP and standardization, and 36 internal configurations. Primary N1 is 0.70094369 for the known optical grid and 0.65103282 for FBCSP. The paired difference is 0.04991087, with bootstrap IC95 [0.01479501, 0.08680927]; Wilcoxon p=0.02446752 and Holm p=0.04893505 retaining the two-hypothesis family with H2 unavailable. It meets the original H1 rule. This does not demonstrate passive captured-network superiority: the optical grid contains a free digital matrix, was designed using known LORO data, and evaluates known subjects. The separate simultaneous-interval sensitivity is positive but is not a substitute for the primary.

The EEGNet complete-fold timing pilot stopped on the real 700 MiB host-RAM floor with NULL timing and no H2 result. Full unittest discovery also stopped on that floor; no complete-suite PASS is claimed. The guide support tests pass 57/57 in two local Python environments and in an automatic Linux GitHub job. The eight selected core modules pass 55 tests in normal/reverse/foreign-cwd selections in both local environments. These are software checks, not independent human replication.

The auxiliary 64-mode Digits mesh has accuracy 0.9492 versus 0.9703 for tuned logistic regression. It remains distinct from the captured geometry. No universal classifier superiority, native resident inference or physical optical energy advantage is asserted.

11. Closest methods and falsifiable contribution

Scene-specific differentiable computation already has strong prior methods, including Dr.Jit (Jakob et al. 2022). Universal interferometer decompositions (Clements et al. 2016) do not establish that the captured geometry reaches the full linear class. In-situ photonic gradients (Pai and others 2023) require actual calibrated physical measurements. Optical energy and photon-limited demonstrations (Hamerly and others 2019; Wang and others 2022) do not make GPU simulation a photonic processor. The present candidate contribution is bounded joint numerical certification at a measured cost; it remains subject to a matched comparison that can refute its utility. Reading coverage and unresolved comparisons are recorded in the related-method matrix.

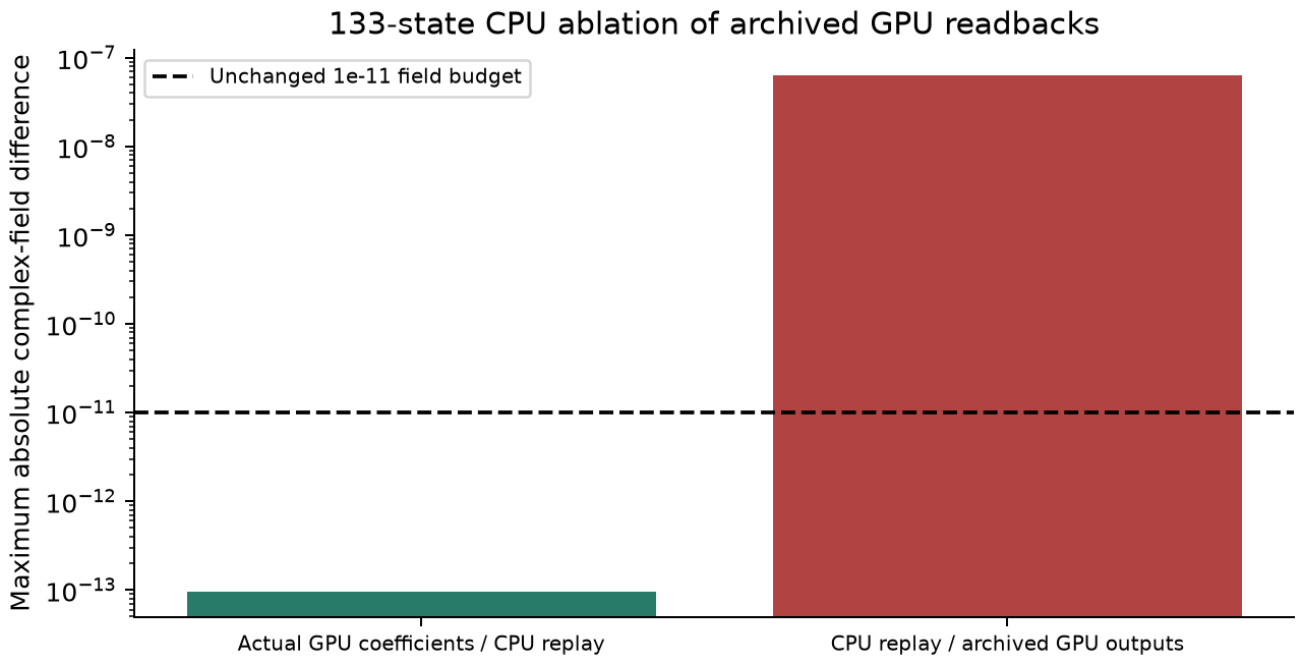


Figure 6: CPU replay of actual historical GPU coefficient readbacks; no new native GPU repair

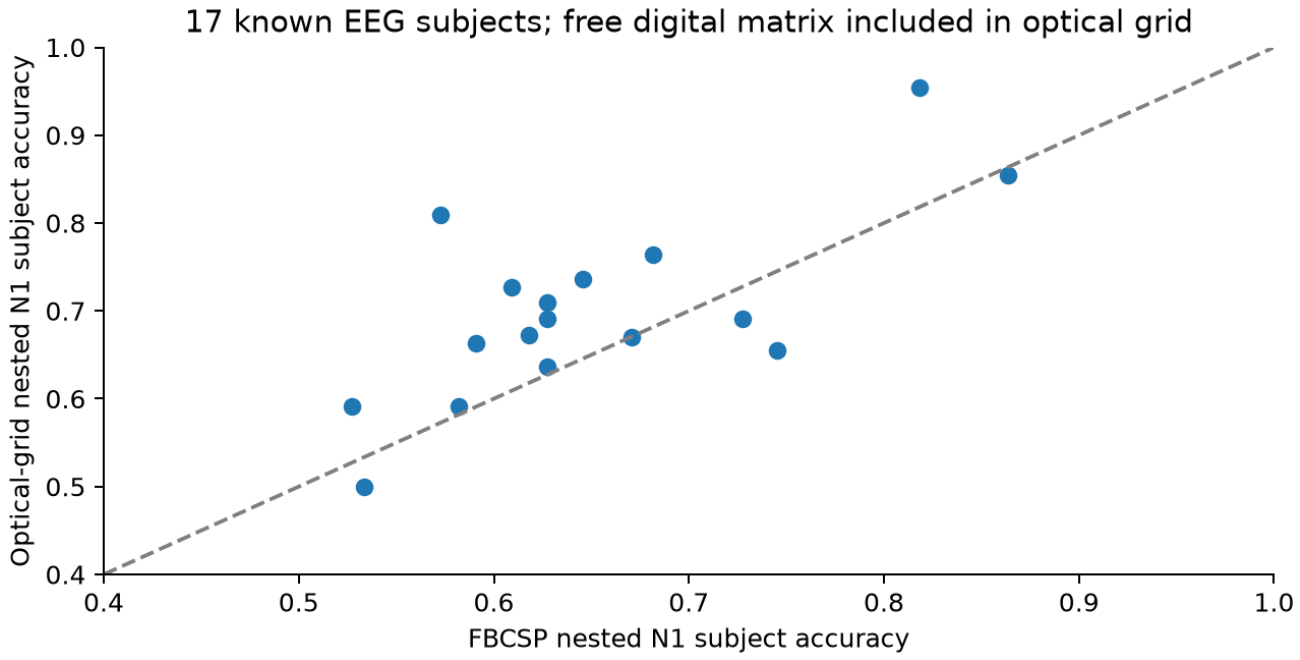


Figure 7: Nested primary N1 accuracies by known EEG subject; the optical grid includes an unconstrained digital matrix

12. Availability, authorship and AI use

The original MIT code, frozen failures and addon 0.1.3 are retained. Codex generated integration, comparator orchestration, diagnostics and manuscript updates; JEV supplied documented advice, not evidence. Real manuscript authors must agree their responsibilities, affiliations and CRediT roles before submission. Raw EEG and derivative labels/logits remain in the owner local workspace until redistribution permission is verified; aggregate results, hashes, code and legitimate-access reproduction commands are public. No new human data, independent specialist review, DOI, journal submission, physical device or adoption record is claimed.

13. Prospective wave confirmation with an independent solver (WAVE-C1)

Section 10 left the diffraction explanation of the P1-7 negative as exploratory. WAVE-C1 tested it prospectively (protocol, pushed to GitHub seven seconds before execution). The reference is a new mesh-free angular-spectrum solver that shares no code with the P1-7 FDTD: exact plane-wave propagation, exact Airy coefficients of a lossless slab for Ez (s) polarization and perfect-conductor mirrors, applied to infinite planar elements in the declared order. Its only idealization beyond the 2D scalar model is the absence of edge diffraction. Because no mesh represents the device, the **same continuous device** (slab n = 2.537430764948469, d = $\lambda/4$, the project's analytic 50/50 design at 45 degrees) is used at every numerical resolution, and the mirror displacement realizes **all 21 phases exactly**. The path models are the frozen P1-7 formula without (M0) and with (M1) free paraxial Gaussian diffraction, using the device's own central-angle coefficients and no fitted parameter.

The solver was self-consistent to 2.7×10^{-15} across spectral sampling and window variants and conserved energy to 3.3×10^{-16}. M1 stayed within 0.0040 of the reference over 21 phases and both ports (budget 0.02, unchanged from P1-7): **confirmed** in the declared regime. M0 erred by 0.0281, **replicating** the P1-7 refutation. The archived FDTD powers agree with the reference carrying the FDTD-calibrated splitter within 0.0048 (budget 0.005), so the FDTD transfer is explained by free diffraction plus its splitter coefficients; with the same refractive index, the discrete FDTD films behave as different devices (difference up to 0.098 at $\lambda/16$). A two-MZI cascade with about 190λ of propagation keeps M1 within 0.0044 while M0 errs by up to 0.107; 21 of its 63 configurations carry no light into the second stage and test nothing, a design weakness found after execution. The confirmation was not blind; the exploratory values were stated as predictions. An external Linux job (different Python, NumPy and SciPy) reran the protocol and reproduced every power within 8.9×10^{-16}.

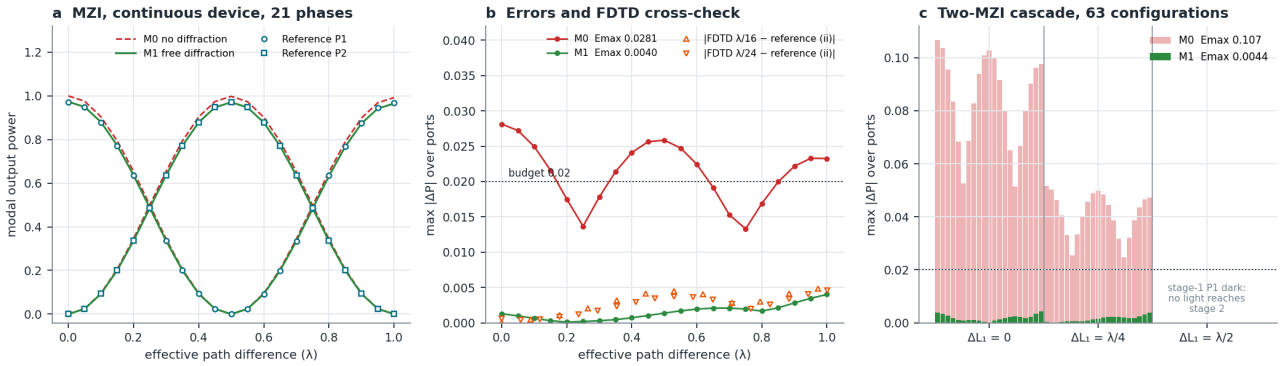


Figure 8: WAVE-C1: continuous device, 21 phases, errors with the FDTD cross-check, and the two-MZI cascade

An independent red-team agent recomputed the MZI by its own direct plane-wave quadrature (1.4×10^{-13}) and refuted one interpretive sentence: the remaining M1 error is the slab's angular dependence (with an angle-independent splitter, solver and M1 differ by about 3×10^{-5}), not the paraxial approximation (red-team report).

The same declared criteria quantify the captured network's regime (results). Its 17,060 terminal paths are 110.7λ to 523.9λ long, while every element is a 4λ square (detectors 3λ), giving a 2.83λ projected aperture. A circular Gaussian beam that is not clipped by more than 10^{-3} per axis has $\zeta = L\lambda/(\pi w^2)$ between 47.7 and 225.7, against $\zeta \leq 0.2857$ for a no-diffraction power error ≤ 0.02. **No path is in regime at λ = 0.1 BU**; all would require a wavelength about 790 times shorter or apertures about 28 times wider relative to the paths. The represented

scalar ray model of the captured scene is therefore a well-defined computational model with certified numerics, but not a faithful wave description of that scene at its stated scale. Physically motivated designs either build diffraction into the model, as diffractive networks do (Lin et al. 2018), or use interferometric meshes whose waveguides confine the modes (Shen et al. 2017); a captured Blender scene would need one of these routes before any physical claim.

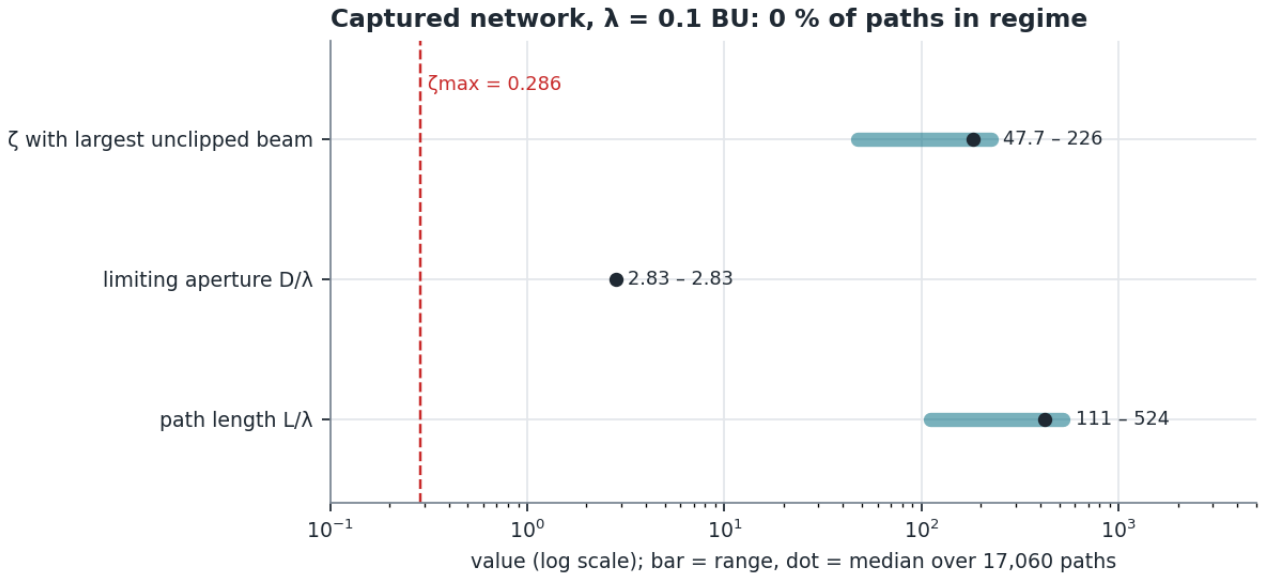


Figure 9: Captured-network regime: path lengths, apertures and the diffraction parameter against the declared limit

14. How much does the captured geometry restrict the decision class? (HA-K1)

With real input each detector is a rank-two quadratic form, so the free linear optical class F_lin is a strict subset of the quadratic class F_quad, and the geometry-realizable class F_geom of the captured scene is a subset of F_lin. The derivative of the three decision detectors' forms with respect to the 16 trained translations has rank 12 at every tested point of the translation box; modulo the decision invariances the local decision dimension is 12 for F_geom, 26 for F_lin and 29 for F_quad. Four translations (cells c03, c13, c23, c33) have exactly zero effect on the decision detectors: a quarter of the historically trained parameters cannot change the classifier.

A preregistered teacher-student test (protocol, results) generated fresh synthetic tasks for 10 seeds with the historical encoding: labels from a random free linear optical map, from random quadratic forms, or from a random setting of the captured geometry. On linear-teacher tasks the free linear student beat the captured-geometry student by +0.051 test accuracy (bootstrap 95 % [0.034, 0.066]; Holm-adjusted one-sided Wilcoxon p = 0.0039; 9 wins, 1 tie): **H-A1 supported**. The optimizer-adequacy control was met (0.971 on realizable tasks), but about 0.02 of the gap reflects optimizer reach rather than representation. The second prediction failed: on quadratic-teacher tasks the quadratic logistic student was slightly less accurate than the linear one (-0.005), so with 400 training inputs the additional quadratic capacity did not become accuracy. Every kept geometry passed a fresh exact topology rebuild. The geometry therefore restricts the linear optical class measurably; an advantage of the approach must come from certification, robustness or cost rather than from expressiveness. Physical training methods such as physics-aware training (Wright et al. 2022) address real devices and are not tested by this represented-model comparison.

15. Compiled operator versus geometry traversal (G1)

Because the represented network is linear at fixed geometry, its 8×5 operator and 16 tangents can be compiled from one traversal of the five basis inputs. The compiled route reproduces the traversal to 6.8×10^{-16} in fields and 5.4×10^{-14} in tangents. The preregistered prediction that it would be at least twice as fast with recompilation

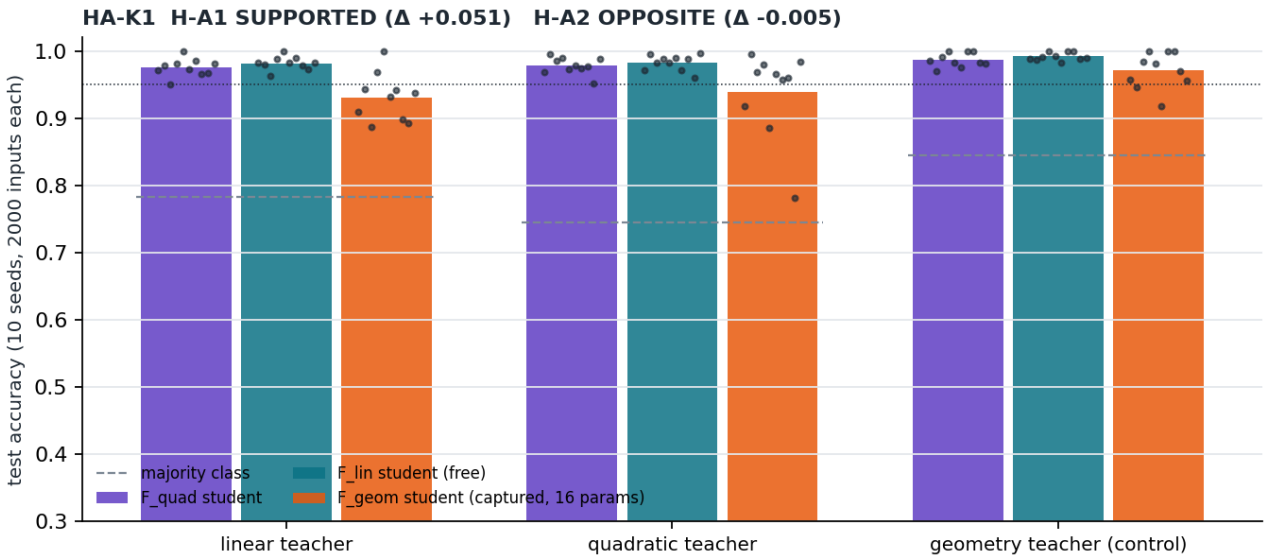


Figure 10: HA-K1: test accuracy of quadratic, free linear and captured-geometry students on three teacher families

charged every call was **refuted** (results): the traversal is already vectorized over the batch, so both routes are dominated by the same per-state loop (ratios 1.14-1.63 for inference, 16.3 only for a training step at batch 4096). With fixed geometry, a cached operator is 29-1583 times faster on this CPU. This is a declared route inside one linear model, not a graphics, RT-core or energy result.

16. Machine-checked certificate lemmas

T1 (decision boxes) was compiled earlier. Lemmas T2-T5 are now proved in core Lean 4.34 without Mathlib (source, record): T2, abs(P_true - P_obs) $\leq$ 2Ae + e^2 for complex fields given by real and imaginary parts; T3, a scaled partial Neumann-tail bound for a sequence with q n(k+1) $\leq$ p n(k), $0 \leq p < q$; T4, vanishing 3×3 minors of $aa^T + bb^T$ (rank ≤ 2 for a real-input detector) and of the principal 2×2 minors of the complex Gram form; T5, no increase of output power for a two-port with $|t|^2 + |r|^2 \leq 1$ and $\mathrm{Re}(t\,\bar{r}) = 0$. `#print axioms` lists only propext, Classical.choice and Quot.sound, and an external Linux job recompiles all five. The statements are over the integers; the red team showed that the T3 hypotheses force an eventually zero integer sequence, so its rational use is per fixed tail by truncation and scaling. Real and complex instantiations, the operator-norm link, infinite sums and the rank-4 bound for a detector difference remain open. These lemmas certify the arithmetic core of the certificates, not that any producer's numbers satisfy their hypotheses.

17. Updated authorship and AI-use note

The WAVE-C1, HA-K1 and G1 protocols, solvers, runners, Lean T2-T5 proofs and this version's new sections were generated by Claude (Anthropic, model claude-opus-5-5) on a separate branch built on Codex's closure branch; an independent Claude verifier agent performed the red-team review. JEV advice is recorded with its provenance and is not evidence. Real authors must still agree on responsibilities and CRediT roles.

References

Clements, William R., Peter C. Humphreys, Benjamin J. Metcalf, W. Steven Kolthammer, and Ian A. Walmsley. 2016. "An Optimal Design for Universal Multiport Interferometers." *Optica*. https://arxiv.org/abs/1603.08788.

Hamerly, and others. 2019. "Large-Scale Optical Neural Networks Based on Photoelectric Multiplication." *Physical Review X*. https://arxiv.org/abs/1812.07614.

Jakob, Wenzel, Sebastien Speierer, Nicolas Roussel, and Delio Vicini. 2022. "Dr.jit: A Just-in-Time Compiler for Differentiable Rendering." *ACM Transactions on Graphics*. https://doi.org/10.1145/3528223.3530099.

Lin, Xing, Yair Rivenson, Nezih T. Yardimci, Muhammed Veli, Yi Luo, Mona Jarrahi, and Aydogan Ozcan. 2018. "All-Optical Machine Learning Using Diffractive Deep Neural Networks." *Science* 361: 1004–8. https://doi.org/10.1126/science.aat8084.

Pai, and others. 2023. "Experimentally Realized in Situ Backpropagation for Deep Learning in Nanophotonic Neural Networks." *Science*. https://arxiv.org/abs/2205.08501.

Shen, Yichen, Nicholas C. Harris, Scott Skirlo, Mihika Prabhu, Tom Baehr-Jones, Michael Hochberg, Xin Sun, et al. 2017. "Deep Learning with Coherent Nanophotonic Circuits." *Nature Photonics* 11: 441–46. https://doi.org/10.1038/nphoton.2017.93.

Wang, and others. 2022. "An Optical Neural Network Using Less Than 1 Photon Per Multiplication." *Nature Communications*. https://arxiv.org/abs/2104.13467.

Wright, Logan G., Tatsuhiro Onodera, Martin M. Stein, Tianyu Wang, Darren T. Schachter, Zoey Hu, and Peter L. McMahon. 2022. "Deep Physical Neural Networks Trained with Backpropagation." *Nature* 601: 549–55. https://doi.org/10.1038/s41586-021-04223-6.